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Roll No

ME-5002-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mechanical Measurement and Control

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Drive the expression for actual flow rate incompressible fluids.
b) Describe the construction and working of a venturimeter.
2. a) What is calibration? Explain static and dynamic calibration.
b) What is meant by measuring range of an instrument? What is the measuring range of a micrometer of 25-30 mm micrometer.
3. a) What is least square regression analysis? When it is useful?
b) Write a note on process capability.
4. a) What are Resistance Strain Gauges? Describe the method to calibrate it.
b) Explain Electromagnetic Technique to measure Angular Velocity.

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5. a) Explain the following terms:
 - i) Transfer Function
 - ii) Signal Flow Graphs
 - iii) Liquid Level Systemb) Discuss the modelling of thermal systems.
6. a) Explain the working of resistance strain gauge.
b) How do you measure torque?
7. a) Differentiate between open loop and closed loop control.
b) Explain the block diagram of closed loop system.
8. a) What is mathematical model? How mathematical equation of a system can be used to derive the transfer function of the system?
b) How the modelling of electrical system can be done?
